

Alexander Iyer

Embedded Systems & IoT Firmware Engineer • 9 Years Experience • MERIQ Index: 70% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Embedded Linux, MQTT, ARM Cortex, FreeRTOS, C/C++, I2C/SPI/UART, Python, PCB Debugging

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Embedded Linux service, reducing latency by 28%.
- Led a squad of 4 engineers delivering scalable microservices processing 18M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 65%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2017 - 2023)

- Designed and maintained MQTT data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using ARM Cortex saving \$13k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Embedded Linux, MQTT, ARM Cortex, FreeRTOS]
Engineered end-to-end production solution leveraging Embedded Linux and MQTT with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [ARM Cortex, FreeRTOS, C/C++, I2C/SPI/UART]
Architected high-resilience system integrating ARM Cortex with 99.9% uptime.

EDUCATION & CREDENTIALS

B.E. in Information Technology

University of Illinois Urbana-Champaign — 2013 - 2017 (Score: 9.04 / 10.0)